FEVER: experimental results

Table 1
LFTO attack parameter setting.

Attack	T1(s)	S(pkt/T)	M(pkt)	Pkt/refresh	Pkt size	T2(s)
A2	[5.5,9]	100	1700	1	64 B	83.5
A4	[6,9.5]	100	1700	1-3	benign-like	103.0
E9P	[2,3]	100	1700	1-3	benign-like	31.0

T2 is the average time from the start of the attack until the flow table overflows without defense.

Table 2
Attack detection results of FEVER.

Attack	Attack detection rate	Attacker identification rate	Detection time(s)	False blocking rate
A2	100 %	100 %	70.9	0 %
A4	100 %	100 %	126.3	0 %
E9P	100 %	100 %	66.9	0 %
No attack	-	-	-	0 %

Table 3
Results of malicious flow identification of LFTO attacks with different parameters.

Attack	Accuracy	Recall	Precision	F1 Score
A2	99.39 %	99.28 %	99.95 %	99.61 %
A4	21.18 %	0.01 %	14.29 %	0.02 %
E9P	10.98 %	0.00 %	0.00 %	0.00 %
Average	43.85 %	33.10 %	38.08 %	33.21 %
Mix	42.69 %	30.66 %	99.88 %	46.92 %

Per flow entry, FEVER's federated detector only. A4 and E9P are left to the volume block.

Table 4
Comparing the outcomes of various classifiers to recognize harmful flows.

Classifier	Accuracy	Recall	Precision	F1 Score	F1 (seen remotely)	F1 (unseen)
FTOP	96.94 %	92.02 %	99.79 %	95.02 %	81.16 %	14.74 %
FTMaster	97.07 %	92.37 %	99.79 %	95.30 %	82.20 %	20.65 %
FloRa	99.01 %	97.50 %	99.82 %	98.62 %	82.68 %	28.46 %
DMS-BTT	97.03 %	92.27 %	99.78 %	95.22 %	82.38 %	19.94 %
FEVER	99.61 %	99.19 %	99.75 %	99.46 %	99.15 %	66.81 %

All classifiers at a 0.1 % false-positive rate on the same data. Seen remotely: attacks seen only by another controller; unseen: an attack no controller was trained on.

Table 5
Comparative experiment results.

Method	Deploy plane	Average Pa	Max Pa	Average flow table usage	Earliest overflow time(s)	Packet-in message	Table-full message
None	None	62.10 %	94.43 %	76.58 %	21	101,854	3,217
FTMaster	Control	45.54 %	82.16 %	60.88 %	31	111,955	1,509
DMS-BTT	Control	41.85 %	78.24 %	58.06 %	40	105,310	271
rDefender*	Control	45.04 %	88.37 %	60.01 %	62	92,780	433
FEVER	Control	11.85 %	67.60 %	31.76 %	-	123,204	0

Mean per run over attacks A2, A4 and E9P; earliest overflow time is counted from the start of the attack (-: the table never overflowed). *3 runs per attack; other methods 5.

Table 5a
Comparative experiment results under A2.

Method	Deploy plane	Average Pa	Max Pa	Average flow table usage	Earliest overflow time(s)	Packet-in message	Table-full message
None	None	58.43 %	94.79 %	75.20 %	75	106,937	3,350
FTMaster	Control	9.98 %	58.10 %	30.21 %	-	158,455	0
DMS-BTT	Control	18.67 %	54.28 %	38.55 %	-	132,215	0
rDefender*	Control	38.64 %	87.58 %	54.85 %	79	104,046	654
FEVER	Control	5.72 %	46.99 %	28.60 %	-	112,814	0

Table 5b
Comparative experiment results under A4.

Method	Deploy plane	Average Pa	Max Pa	Average flow table usage	Earliest overflow time(s)	Packet-in message	Table-full message
None	None	54.77 %	91.38 %	70.94 %	83	99,238	1,828
FTMaster	Control	53.36 %	91.42 %	69.06 %	104	90,553	1,307
DMS-BTT	Control	47.58 %	86.21 %	64.63 %	132	96,989	245
rDefender*	Control	44.71 %	86.92 %	62.90 %	85	100,985	364
FEVER	Control	16.38 %	74.22 %	35.99 %	-	119,845	0

Table 5c
Comparative experiment results under E9P.

Method	Deploy plane	Average Pa	Max Pa	Average flow table usage	Earliest overflow time(s)	Packet-in message	Table-full message
None	None	73.10 %	97.13 %	83.60 %	21	99,386	4,472
FTMaster	Control	73.29 %	96.97 %	83.38 %	31	86,856	3,219
DMS-BTT	Control	59.30 %	94.23 %	71.00 %	40	86,725	568
rDefender*	Control	51.78 %	90.61 %	62.28 %	62	73,310	280
FEVER	Control	13.44 %	81.61 %	30.70 %	-	136,954	0

Table 6
Impact on benign users.

Method	Benign flows harmed	Lost first packets	New connections succeeding	Decision time(ms)	Attackers blocked
None	10.13 %	28.2 %	25.2 %	30	0/30
FTMaster	4.80 %	27.8 %	23.6 %	72	0/30
DMS-BTT	1.71 %	32.6 %	23.4 %	65	0/30
rDefender*	1.83 %	36.4 %	20.8 %	26	0/18
FEVER	0.01 %	9.6 %	37.3 %	11	30/30

*3 runs per attack; other methods 5.